

WTW158 Calculus

Class Test 6 (Mo)

8 - 12 May 2023

Time: 15 min

8

Surname:

Initials:

Student number:

Group:

Use a pen to complete the test. The test is printed on both sides of this paper.

QUESTION 1 [1,1]

Find the derivative and do not simplify your answer.

(1.1) $g(t) = \sinh(\cos(t-1))$.

$$g'(t) = \cosh(\cos(t-1)) \times (-\sin(t-1))$$

(1.2) $g(t) = \ln \cosh(\sin(t))$

$$g'(t) = \frac{1}{\cosh(\sin(t))} \times \sinh(\sin(t)) \times \cos t$$

QUESTION 2 [1 mark]

Find the critical numbers of $f(t) = 2t e^t$ on $[-3, 3]$.

$$\begin{aligned} f'(t) &= 2e^t + 2te^t \\ &= 2e^t(1+t) = 0 \\ \Rightarrow t &= -1. \end{aligned}$$

Critical number: $t = -1$.

QUESTION 3 [2 marks]

Indicate whether the statement is True or False:

(3.1) If $f'(c) = 0$, then f has a local maximum or minimum in c , provided that c is in the domain of f .

TRUE / FALSE

(3.2) A function f , defined on a closed interval I , will always attain an absolute maximum value at a number in the closed interval I .

TRUE / FALSE

QUESTION 4 [1 mark]

Complete:

Suppose c is a number in the domain D of f . Then $f(c)$ is the absolute maximum value of f if

$$f(c) \geq f(x) \text{ for all } x \in D.$$

QUESTION 5 [2 marks]

Find the derivative of $y = \sin(x)^{\tan(x)}$

$$\ln y = \ln (\sin x^{\tan x}) = \tan x \ln \sin x$$

$$\frac{d}{dx}(\ln y) = \frac{d}{dx}(\tan x \ln(\sin x))$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \sec^2 x \ln(\sin x) + \tan x \frac{\cos x}{\sin x}$$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \sin x^{\tan x} \left[\sec^2 x \ln(\sin x) + \tan x \frac{\cos x}{\sin x} \right] \\ &= \sin x^{\tan x} \left[\sec^2 x \ln \sin x + 1 \right] \end{aligned}$$

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QUESTION 1 [1,1]

Find the derivative and do not simplify your answer.

(1.1) $g(t) = \sinh(\cos(t) - 1)$

$$g'(t) = \cosh(\cos t - 1) \times (-\sin t)$$

(1.2) $g(t) = \ln \sinh(\cos(t))$

$$g'(t) = \frac{1}{\sinh(\cos t)} \times \cosh(\cos t) \times (-\sin t)$$

QUESTION 2 [1 mark]

Find the critical numbers of $f(t) = 2t e^t$ on $[-3, 3]$.

$$\begin{aligned} f'(t) &= 2e^t + 2te^t \\ &= 2e^t(1+t) \end{aligned}$$

$$f'(t) = 0 \Rightarrow 2e^t(1+t) = 0$$

$$\Rightarrow 1+t = 0$$

$$\Rightarrow t = -1$$

$$\Rightarrow \text{Critical number: } t = -1.$$

QUESTION 3 [2 marks]

Indicate whether the statement is True or False:

(3.1) If $f'(d)$ does not exist and d is in the domain of f , then d is a critical number of f .

TRUE / FALSE

(3.2) A function f , defined on a closed interval I , will always attain an absolute maximum value at a number in the closed interval I .

TRUE / FALSE

QUESTION 4 [1 mark]

Complete:

Suppose c is a number in the domain D of f . Then $f(c)$ is the absolute maximum value of f if

$$f(c) \geq f(x) \text{ for all } x \in D.$$

QUESTION 5 [2 marks]

Find the derivative of $y = \sin(x)^{\cos(x)}$

Consider $\ln y = \ln(\sin x^{\cos x})$
 $= \cos x \ln(\sin x)$

$$\Rightarrow \frac{d}{dx} \ln y = \frac{d}{dx} (\cos x \ln(\sin x))$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = (-\sin x) \ln \sin x + \cos x \times \frac{1}{\sin x} \times \cos x$$

$$\Rightarrow \frac{dy}{dx} = y \left[-\sin x \ln \sin x + \frac{\cos^2 x}{\sin x} \right]$$

$$= \sin x^{\cos x} \left[-\sin x \ln(\sin x) + \frac{\cos^2 x}{\sin x} \right]$$

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QUESTION 1 [1,1]

Find the derivative and do not simplify your answer.

(1.1) $g(t) = \sin(\cosh(t) - 1)$

$$g'(t) = \cos(\cosh(t) - 1) \times \sinh(t)$$

(1.2) $g(t) = \ln(\sinh(\cos(t)))$

$$g'(t) = \frac{1}{\sinh(\cos t)} \times \cosh(\cos t) \times (-\sin t)$$

QUESTION 2 [2 marks]

Find the derivative of $y = \tan(x)^{\sin(x)}$

$$\ln y = \ln(\tan^{\sin x}) = \sin x \ln(\tan x)$$

$$\Rightarrow \frac{d}{dx}(\ln y) = \frac{d}{dx}(\sin x \ln(\tan x))$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \cos x \ln(\tan x) + \sin x \frac{\sec^2 x}{\tan x}$$

$$\Rightarrow \frac{dy}{dx} = (\tan x)^{\sin x} \left[\cos x \ln(\tan x) + \sin x \frac{\sec^2 x}{\tan x} \right]$$

QUESTION 3 [2 marks]

Indicate whether the statement is True or False:

(3.1) If $f'(d) = 0$ and d is in the domain of f , then d is a critical number of f .

TRUE / FALSE

(3.2) A function f , continuous on a closed interval I , will always attain an absolute maximum value at a number in the closed interval I .

TRUE / FALSE

QUESTION 4 [1 mark]

Complete:

Suppose c is a number in the domain D of f . Then $f(c)$ is the local minimum value of f if

$f(c) \leq f(x)$ for all x in a vicinity of c .

QUESTION 5 [1 mark]

Find the critical numbers of $f(x) = x^3 - 12x + 6$ on $[0, 10]$.

$$f'(x) = 3x^2 - 12$$

$$f'(x) = 0 \Rightarrow 3(x^2 - 4) = 0$$

$$\Rightarrow 3(x-2)(x+2) = 0$$

$$\Rightarrow x = 2 \text{ or } x = -2$$

Critical numbers must lie in $[0, 10]$

$\Rightarrow$ Critical nr of f is: $x = 2$.

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QUESTION 1 [1,1]

Find the derivative and do not simplify your answer.

(1.1) $g(t) = \cos(\cosh(t) - 1)$

$$g'(t) = -\sin(\cosh(t) - 1) \times \sinh(t)$$

(1.2) $g(t) = \ln(\sinh(\cos(t)))$

$$g'(t) = \frac{1}{\sinh(\cos(t))} \times \cosh(\cos(t)) \times (-\sin(t))$$

QUESTION 2 [2 marks]

Find the derivative of $y = \tan(x)^{\sin(x)}$

$$\ln y = \ln(\tan^{\sin x}) = \sin x \ln(\tan x)$$

$$\frac{d}{dx}(\ln y) = \frac{d}{dx}(\sin x \ln(\tan x))$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \cos x \ln(\tan x) + \sin x \times \frac{\sec^2 x}{\tan x}$$

$$\Rightarrow \frac{dy}{dx} = \tan^{\sin x} \left(\cos x \ln(\tan x) + \frac{\sin x \sec^2 x}{\tan x} \right)$$

QUESTION 3 [2 marks]

Indicate whether the statement is True or False:

(3.1) If $f'(d)$ does not exist, then d is a critical number of f , provided d is in the domain of f .

TRUE FALSE

(3.2) A function f , continuous on a closed interval I , will always attain an absolute maximum value at a number in the closed interval I .

TRUE FALSE

QUESTION 4 [1 mark]

Complete:

Suppose c is a number in the domain D of f . Then $f(c)$ is the local minimum value of f if

$f(c) \leq f(x)$ for all x in a vicinity of c

QUESTION 5 [1 mark]

Find the critical numbers of $f(x) = x^3 - 12x + 6$ on $[0, 10]$.

$$f'(x) = 3x^2 - 12$$

$$f'(x) = 0 \Rightarrow 3(x^2 - 4) = 0$$

$$\Rightarrow 3(x-2)(x+2) = 0$$

$$\Rightarrow x = 2 \text{ or } x = -2$$

Critical numbers must lie in $[0, 10]$

$\Rightarrow x = 2$ is a critical nr of f .